

When Do Failures Reveal Spurious Structure?

A Study of Label-Free Latent-Risk Editing in Frozen Vision-Language Models

Anonymous submission

Abstract

Errors of frozen vision-language models (VLMs) can concentrate in groups that are unobserved during adaptation. We ask whether task failures alone expose an editable latent-risk subspace without group or sensitive-attribute labels. LATENTRISK-VLM converts task-labeled calibration margins into risk labels, removes class-level semantics, and estimates class-conditional risk directions by residual mean contrast or iterative linear probes. The subspace is defined operationally by failure predictiveness, weak semantic coupling, and suppressibility; it is not a discovered sensitive attribute. At inference, every candidate class receives its own correction, so the unknown test label never selects an edit. Experiments with frozen CLIP ViT-L/14 show a conditional rather than universal benefit. On Waterbirds, a task-label-only rank-1 editor raises worst-group accuracy (WGA) from 33.6 to 60.5 while average accuracy falls from 83.7 to 78.4. On CelebA, rank-1 editing instead lowers WGA from 71.5 to 54.4 despite raising average accuracy from 78.2 to 91.7. Attribute-labeled validation changes rank-4 WGA from 52.2 to 82.2, exposing a major selection bottleneck, while an attribute-selected CnC baseline reaches 85.3. In retrieval, FairFace gender MaxSkew@100 decreases from 0.355 to 0.323 as Flickr30K Recall@10 decreases from 92.2 to 88.2. Failure-derived directions are therefore useful weak supervision when failure and group shift align, but neither risk separability nor task accuracy guarantees group robustness.

Introduction

Frozen vision-language models (VLMs) provide transferable image-text representations for classification and retrieval (Radford et al. 2021). Their zero-shot interface, however, does not remove biases inherited from web-scale pretraining. Errors can concentrate on demographic groups (Agarwal et al. 2021; Hall et al. 2023), but also on backgrounds, acquisition conditions, pose, or other spurious factors. Standard average accuracy can conceal these failures.

Most post-hoc VLM debiasing methods assume that the relevant attribute is known. They encode named concepts in prompts (Chuang et al. 2023), use attribute-labeled reference images (Gerych et al. 2024), or estimate attribute-predictive

subspaces (Zhao et al. 2026). This is appropriate when the attribute is available and is the actual source of harm. In many deployments, however, group labels are unavailable, restricted, incomplete, or simply miss the factor driving errors. What remains observable in deployment is the model’s pattern of errors and low-confidence predictions.

We therefore ask: *can low classification margins—where the correct-class score is close to that of the strongest competing class—and prediction errors reveal a representation direction whose removal reduces hidden-group failure?* This question differs from discovering a demographic attribute. A failure-predictive direction may encode background, image quality, an intersection of factors, or useful class evidence. Consequently, editing it may improve worst-group reliability, do nothing, or make performance less balanced. A credible method must expose this distinction rather than equating any failure direction with a fairness direction.

We call a low-dimensional subspace a *latent risk subspace* when it is predictive of within-class calibration failures, weakly coupled to class semantics, and suppressible without destroying the task score. This is an operational representation of error concentration, not a recovered gender, race, age, or other sensitive attribute. If it overlaps the factor defining a worst-performing group, suppressing it can indirectly improve worst-group performance; if it does not, editing can be harmful.

We introduce LATENTRISK-VLM, a post-hoc framework for frozen VLM embeddings. Given a task-labeled calibration set, we derive risk labels from zero-shot margins and remove class prototypes and text-aligned evidence. We estimate a class-conditional risk direction either by contrasting high- and low-risk residual means or by fitting iterative linear probes inspired by INLP (Ravfogel et al. 2020). Test labels are not used: each candidate class supplies its own prototype and subspace, producing candidate-wise corrected scores. All selection in the primary protocol uses task labels only; sensitive attributes are read after prediction solely for group metrics.

The completed experiments give a deliberately mixed answer. On Waterbirds, rank-1 editing improves WGA by 26.9 points without place labels, although it loses 5.3 points of average accuracy. On CelebA Blond Hair, rank-1 editing increases average accuracy by 13.5 points but *reduces* WGA by 17.1 points. Attribute-labeled selection changes rank-4

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CelebA WGA from 52.2 to 82.2, showing that discovering a risk direction and selecting a group-robust operating point are separate problems; CnC with attribute-labeled validation reaches 85.3. A FairFace/Flickr30K retrieval audit likewise finds a modest skew reduction accompanied by a 4.0-point Recall@10 loss.

Our contributions are threefold. First, we formulate latent-risk editing using a calibration set with task labels but no group attributes. At test time, because the true class is unknown, the method edits and scores each candidate class separately rather than selecting an editor using the ground-truth label. Second, we develop a rank-1 residual-contrast editor and a probe-based multidirectional extension, while keeping both VLM encoders frozen. Third, experiments across classification and retrieval reveal two conditions for successful editing: model failures must be correlated with the factor defining the hidden groups, and a group-label-free validation criterion must be able to select appropriate editing parameters. The contrasting results on Waterbirds and CelebA are therefore central to our conclusion: failures provide useful weak supervision, but they do not consistently identify hidden groups.

Related Work

VLM bias measurement and post-hoc editing. Audits have documented demographic and stereotype-related disparities in VLM classification and retrieval (Agarwal et al. 2021; Hall et al. 2023; Zhou, Lai, and Jiang 2022; Hamidieh et al. 2024). Prompt and projection methods remove directions specified by biased prompt pairs (Berg et al. 2022; Chuang et al. 2023); BEND-VLM uses attribute-labeled references for query-specific correction (Gerych et al. 2024); SANER learns a societal-attribute neutralizer from a predefined lexicon without per-image attribute labels (Hirota et al. 2025); and recent work treats bias as an attribute-predictive subspace (Jung, Jang, and Wang 2024; Zhao et al. 2026). These methods start from an attribute or concept. We instead start from within-class failure and test post hoc whether the resulting editor improves group metrics.

Robustness without training-group labels. Group DRO directly optimizes worst-group risk when group annotations are available (Sagawa et al. 2020). JTT uses early errors to up-weight hard examples (Liu et al. 2021), EIL infers environments from loss (Creager, Jacobsen, and Zemel 2021), and last-layer methods improve robustness with limited group supervision (Kirichenko, Izmailov, and Wilson 2023; Qiu et al. 2023). Correct-N-Contrast forms contrastive sets from error-derived groups (Zhang et al. 2022). Our setting is more restrictive in model access: image and text encoders remain frozen, and intervention occurs in their shared embedding space. It is also diagnostically different: we do not assume that errors recover the true groups, and our CelebA result demonstrates the cost of that assumption.

Representation subspaces. INLP iteratively removes linearly decodable information (Ravfogel et al. 2020), while recent VLM work argues that social bias is distributed across subspaces rather than individual coordinates (Zhao et al.

2026). We evaluate both a transparent mean-contrast direction and iterative risk probes. On Waterbirds, mean contrast outperforms the rank-4 construction; on CelebA, even a highly risk-separable rank-1 probe harms WGA. Our focus is therefore not whether risk is decodable, but whether suppressing it preserves task semantics and improves held-out groups.

LatentRisk-VLM

Problem Setup

Let f_I and f_T be frozen image and text encoders. For image x_i and class prompt p_c , the unit-normalized embeddings are $z_i = f_I(x_i) \in \mathbb{R}^d$ and $t_c = f_T(p_c) \in \mathbb{R}^d$. Frozen zero-shot prediction is

$$\hat{y}_i^{\text{zs}} = \arg \max_{c \in \mathcal{Y}} z_i^\top t_c. \quad (1)$$

We observe a calibration set $D_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^n$ with task labels but no group attributes. A group attribute a_i (place in Waterbirds or gender in CelebA) may be revealed only after model selection to compute group accuracy. The primary protocol forbids a_i during direction discovery, intervention, hyperparameter selection, and test-time scoring.

The calibration margin and risk are

$$m_i = z_i^\top t_{y_i} - \max_{c \neq y_i} z_i^\top t_c, \quad (2)$$

$$\rho_i = \sigma(-m_i/\tau), \quad h_i = \mathbb{I}[\rho_i \geq \eta]. \quad (3)$$

A negative margin is a zero-shot error, so with the measured setting $\eta = 0.5$, h_i separates errors from correct predictions, where $h_i = 1$ indicates an incorrect zero-shot prediction and $h_i = 0$ indicates a correct zero-shot prediction. We seek a class-conditional, low-rank representation U_c with three properties: it predicts h_i after class semantics are removed, has weak coupling to the class score, and can be suppressed without erasing the score needed for class c . These properties define a latent risk subspace; they do not imply that U_c encodes a named sensitive attribute.

At test time y is unknown. ‘‘Class-conditional’’ therefore cannot mean selecting U_y with the ground-truth label. For every candidate c , the method constructs a corrected embedding $\hat{z}^{(c)}$ and score $s_c(x)$, then predicts $\arg \max_c s_c(x)$. This candidate-wise rule is part of the problem definition and prevents test-label leakage.

Figure 1 shows the complete pipeline. The backbone is used only to encode images and text; all subsequent operations occur in its normalized embedding space.

Removing Class-Level Semantics

A risk estimator applied directly to z_i may rely on ordinary class-discriminative evidence rather than features genuinely associated with classification error. For each class c , we therefore compute a class prototype, defined as the mean embedding of its calibration samples:

$$\mu_c = \frac{1}{|D_c|} \sum_{i: y_i=c} z_i, \quad (4)$$

and form a residual orthogonal to the class text embedding:

$$r_i = z_i - \mu_{y_i}, \quad \tilde{z}_i = r_i - t_{y_i} t_{y_i}^\top r_i. \quad (5)$$

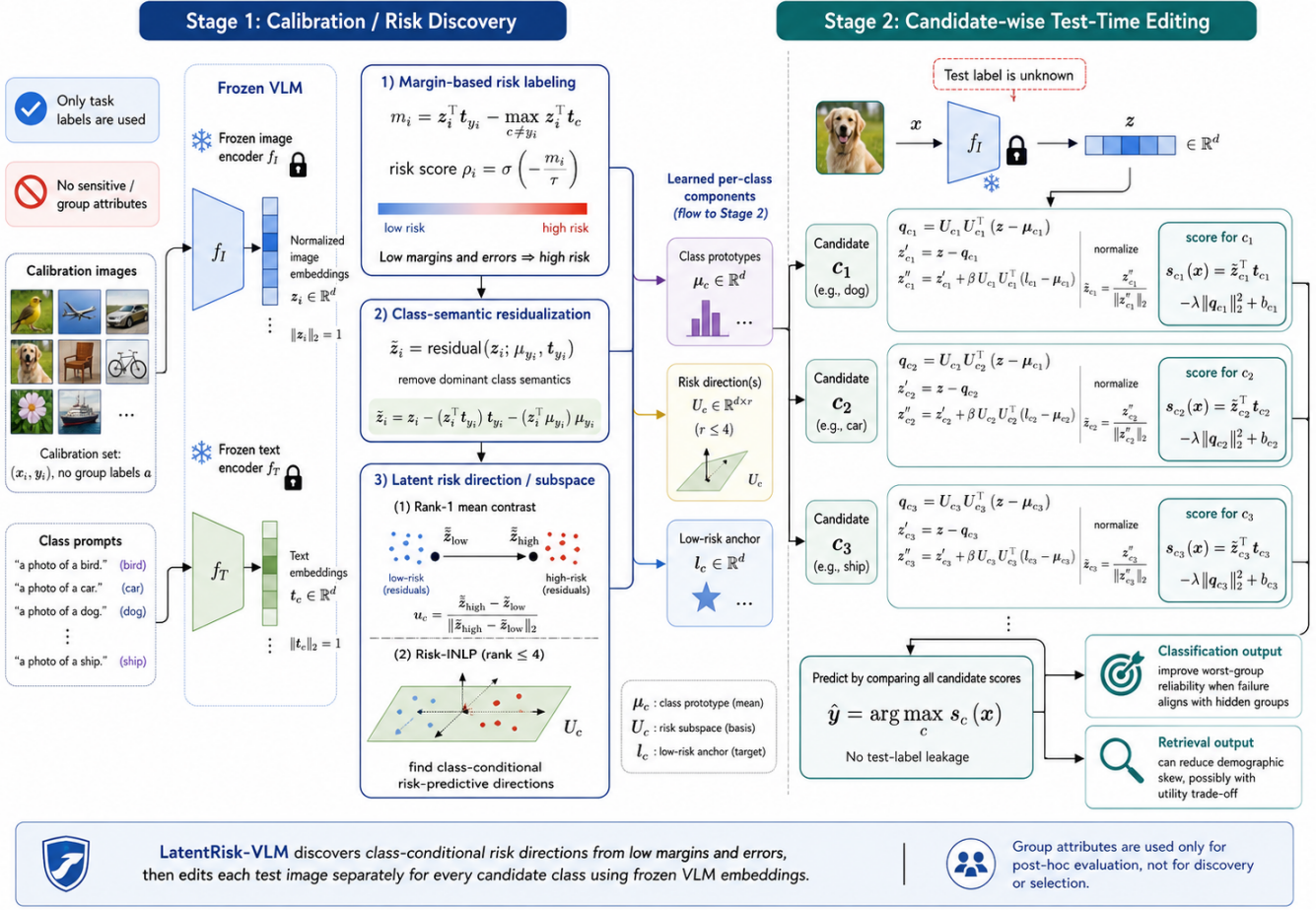


Figure 1: LATENTRISK-VLM. Risk directions are estimated from high- and low-risk residuals using mean contrast or iterative probes. At test time every candidate class is scored with its own correction; no test label selects the edit.

Subtracting the class prototype restricts risk-direction discovery to within-class variation, while the second operation reduces the direct coupling between the representation and the corresponding class-prompt direction. However, these operations do not guarantee that all class information has been completely removed.

Unified design objective. We aim to identify a risk direction, or risk subspace, U_c that captures error-related information while containing as little class-semantic information as possible, such that suppressing it does not impair the classification capability. A compact specification is

$$\min_{U_c, \hat{z}^{(c)}} \mathcal{L}_{\text{risk}}(U_c; \tilde{Z}_c, h_c) + \alpha \|t_c^\top U_c\|_2^2 + \lambda \|U_c U_c^\top (z - \mu_c)\|_2^2 - \hat{z}^{(c)\top} t_c. \quad (6)$$

Here, $U_c^\top U_c = I$, $\text{rank}(U_c) \leq r$, $\hat{z}^{(c)} = \text{Edit}(z; U_c, \ell_c)$.

The first term encourages the learned U_c to predict calibration risk. The second discourages coupling between the risk subspace and the class semantics. A large value of $\|t_c^\top U_c\|_2^2$ indicates that the learned risk directions are strongly

aligned with the class text embedding, suggesting that the model may be capturing class-discriminative information rather than risk-related information. The third term penalizes the candidate's activation in the risk subspace. Specifically, $U_c U_c^\top (z - \mu_c)$ projects the class-centered feature onto the risk subspace. A large norm indicates that the candidate contains a strong risk component; therefore, the objective encourages this activation to be reduced. The fourth term preserves task semantics by encouraging the edited image embedding $\hat{z}^{(c)}$ to remain highly similar to the class text embedding t_c .

Regarding the constraints, $U_c^\top U_c = I$ requires the columns of U_c to be orthonormal. In other words, for $U_c = [u_1, u_2, \dots, u_r]$, we have $u_i^\top u_j = 0$ for $i \neq j$ and $u_i^\top u_i = 1$. The constraint $\text{rank}(U_c) \leq r$ restricts the search to a low-dimensional risk subspace. Finally, $\hat{z}^{(c)} = \text{Edit}(z; U_c, \ell_c)$, where Edit denotes the normalized candidate-wise projection and reinjection defined in Equations 8–10: the risk component is first removed by projection, after which neutral information is reinjected.

This formulation is a design objective rather than an addi-

tional jointly optimized loss. In practice, class residualization reduces semantic coupling, mean contrast or Risk-INLP estimates the risk subspace, projection implements suppressibility, and neutral reinjection together with the semantic score preserves utility. The scoring grid determines the operating point, and α is not tuned separately.

Estimating Risk Directions

Let $H_c = \{i : y_i = c, h_i = 1\}$ denote the set of high-risk or misclassified samples from class c , and let $L_c = \{i : y_i = c, h_i = 0\}$ denote the corresponding low-risk or correctly classified samples. We estimate a class-specific risk direction by contrasting the mean residualized embeddings of these two groups:

$$u_c = \frac{\bar{z}_{H_c} - \bar{z}_{L_c}}{\|\bar{z}_{H_c} - \bar{z}_{L_c}\|_2}, \quad \bar{z}_S = \frac{1}{|S|} \sum_{i \in S} \tilde{z}_i. \quad (7)$$

Here, $\bar{z}_{H_c}$ and $\bar{z}_{L_c}$ are the centroids of the high-risk and low-risk samples, respectively, in the residualized feature space. Their difference therefore points from the average representation of successful predictions toward that of failures. Normalizing this difference removes its scale and yields a unit vector that captures only the dominant direction of risk-related variation within class c .

Thus, $U_c = u_c \in \mathbb{R}^{d \times 1}$ defines a one-dimensional, class-conditional risk subspace. For a residualized embedding $\tilde{z}_i$, the projection $u_c^\top \tilde{z}_i$ measures its activation along this direction, with larger values indicating greater alignment with the feature pattern associated with failures. The mean-contrast estimator provides the simplest linear description of how failures differ from successes after class residualization. Since it is constructed directly from the separation between high-risk and low-risk samples, it is risk-predictive by design. However, because no demographic or sensitive-attribute annotations are used, the resulting direction should be interpreted only as a latent risk direction rather than as a named demographic direction.

Probe-based directions. Alternatively, we fit a linear logistic probe to h_i within each class. Its normalized weight becomes $u_{c,1}$; the residuals are projected into its null space, and the process repeats up to rank four or until probe accuracy falls below 0.55. QR orthogonalization yields $U_c = [u_{c,1}, \dots, u_{c,r}]$. This Risk-INLP estimator follows the iterative removal principle of Ravfogel et al. (2020). Rank one tests a single discriminative direction, while rank four tests whether residual risk remains distributed after successive projections.

Candidate-Wise Representation Editing

For a test embedding z and candidate class c , define its residual risk component $q_c = U_c U_c^\top (z - \mu_c)$. Full projection gives

$$z'^{(c)} = z - q_c. \quad (8)$$

Projection may move an embedding away from regions represented by low-risk calibration examples. Let ℓ_c be the mean original embedding over L_c . We reinsert half of its risk-subspace component,

$$z''^{(c)} = z'^{(c)} + \beta U_c U_c^\top (\ell_c - \mu_c), \quad \beta = 0.5, \quad (9)$$

and normalize $\hat{z}^{(c)} = z''^{(c)} / \|z''^{(c)}\|_2$. The candidate score is

$$s_c(x) = \hat{z}^{(c)\top} t_c - \lambda \|q_c\|_2^2 + b_c. \quad (10)$$

The first term retains CLIP semantic matching, the second penalizes candidate-specific risk activation, and b_c is a class intercept. Prediction compares Equation 10 over all c ; it never applies a correction chosen by the ground-truth test class.

Task-Label-Only Selection

We choose λ and b_c on calibration data using accuracy (CelebA) or macro class accuracy (Waterbirds), with overall accuracy and smaller intervention parameters as tie breakers. Subspace rank, projection strength, and β are fixed before this grid search. The primary selection score depends on y but not a . For diagnosis, we rerun the same search using validation WGA over (y, a) and label the resulting row *attribute oracle*. Comparing the two selectors isolates a practical question: even if error-derived directions contain group-relevant information, can an attribute-free criterion choose a useful operating point?

Complexity. Using cached embeddings of dimension d , rank-1 discovery requires one pass over n calibration samples, $O(nd)$ time and $O(nd)$ storage. Candidate-wise inference costs $O(|\mathcal{Y}|d)$ per image beyond the encoder call. The rank- r extension increases projection cost to $O(|\mathcal{Y}|dr)$ and fits at most r linear probes per class. No variant adds a VLM forward pass or modifies backbone parameters.

Retrieval Adaptation

For Flickr30K calibration captions, risk is the negative margin between the paired image and the hardest nonmatching image. Paired image embeddings are centered by their global prototype and residualized against their caption embeddings. Splitting at the median margin yields one high–low risk direction. The same projection and reinjection edit every gallery image, while λ is selected by validation text-to-image Recall@10. The fitted editor is then evaluated unchanged on Flickr30K test retrieval and on FairFace demographic skew; FairFace gender labels never enter fitting or ranking.

Experimental Protocol

Backbone and reporting. All experiments use 768-dimensional, normalized embeddings from the Transformers CLIP ViT-L/14 implementation; both encoders are frozen. We report one deterministic run for each configuration. Results are measured adaptations to this common backbone, not numbers copied from the baseline papers.

Classification datasets. Waterbirds (Sagawa et al. 2020) uses the official validation split (1,199 images) for risk discovery and parameter selection and the test split (5,794 images) for evaluation. Fixed prompts are “a photo of a landbird” and “a photo of a waterbird.” Groups cross bird type with place. CelebA (Liu et al. 2015) uses Blond Hair as the task and Male as the evaluation attribute. Its validation split (19,867 images) calibrates LATENTRISK-VLM; its training split is available to fitted baselines; and its test split (19,962 images) is evaluated once. Prompts describe a person with or without blond hair.

Retrieval datasets. FairFace (Karkkainen and Joo 2021) supplies a 10,954-image validation gallery and the 25 gender-stereotype queries released with BEND-VLM. Its 86,744-image training split is used only as the gender-labeled reference bank for BEND-VLM. We report text-to-image retrieval on the Flickr30K (Young et al. 2014) Karpathy test split (Karpathy and Fei-Fei 2015): 5,000 captions and 1,000 images. The 1,014-image validation split fits the LatentRisk retrieval editor. SANER is trained for five epochs on 124,026 person-related caption pairs from 27,576 disjoint Flickr30K training images.

Supervision accounting. We separate three classification settings. *No example labels* includes frozen CLIP and prompt ensembling. *Task labels only* includes ERM, JTT-style calibration, and the primary LATENTRISK-VLM variants. *Attribute-supervised or selected* includes any method that uses a for adaptation, references, checkpoint choice, or validation-WGA selection: SPD, BEND-VLM, CFR, CnC, GIC, AFR, and our attribute oracle. This conservative accounting prevents validation access to a from being presented as group-label-free.

Baselines. Classification evaluates frozen CLIP, prompt ensembling, ERM, JTT (Liu et al. 2021), AFR (Qiu et al. 2023), CnC (Zhang et al. 2022), BEND-VLM (Gerych et al. 2024), SPD (Zhao et al. 2026), CFR (You et al. 2024), and a GIC adaptation. Retrieval evaluates frozen CLIP, projection prompts (Chuang et al. 2023), BEND-VLM, SANER (Hirota et al. 2025), and LATENTRISK-VLM. Where a reference method originally trains a different backbone, we retain its central objective and selection rule but operate on CLIP ViT-L/14 features or lightweight adapters. This creates a controlled frozen-backbone comparison, not an exact reproduction of each paper’s original setting.

Hyperparameters. Classification uses $\tau = 1$, $\eta = 0.5$, full projection, and $\beta = 0.5$. The measured rank-1 configuration uses mean contrast on Waterbirds and one logistic probe on CelebA; rank four uses iterative probes with $C = 1$ and a 0.55 training-accuracy stopping threshold. Waterbirds searches 101 values of $\lambda \in [0, 5]$ and 61 intercepts in $[-0.03, 0.03]$; CelebA searches 51 values in $[0.05, 0.30]$ over the same intercept grid. Retrieval searches 61 values of $\lambda \in [0, 0.3]$ and selects 0.1. The Waterbirds ablation changes one component at a time under task-label-only macro-accuracy selection.

Metrics. Classification reports average accuracy and WGA over the four (y, a) groups. We also discuss their difference as the average–worst gap. Retrieval reports text-to-image Recall@10 and mean gender MaxSkew@100 over the 25 FairFace queries. MaxSkew compares retrieved gender proportions with a uniform target; lower is better. Group attributes are accessed only after prediction except in rows explicitly marked attribute-supervised.

Measured Results and Analysis

Classification

Table 1 reports the measured classification results. On Waterbirds, rank-1 LATENTRISK-VLM is strongest among task-label-only methods at 60.5 WGA, improving frozen CLIP by 26.9 points while reducing average accuracy by 5.3 points. Among attribute-supervised or attribute-selected methods, our oracle reaches 64.5 WGA and AFR reaches 61.8. Thus the Waterbirds result supports failure-derived editing when the nuisance is unnamed, while exposing an explicit robustness–utility tradeoff.

CelebA reverses the primary result. Rank-1 editing raises average accuracy from 78.2 to 91.7 but lowers WGA from 71.5 to 54.4; rank four reaches 52.2 WGA. Changing only the rank-4 selection objective from validation accuracy to validation WGA raises test WGA by 30.0 points, from 52.2 to 82.2, while lowering average accuracy to 85.7. This oracle does not establish attribute-free robustness: it uses Male during selection and remains below the 85.3 WGA of attribute-selected CnC. The result instead isolates selection as a bottleneck after risk-subspace discovery.

Retrieval Skew under Demographic Audit

Table 2 shows that LATENTRISK-VLM reduces FairFace MaxSkew from 0.355 to 0.323 while Flickr30K R@10 falls by 4.0 points. Projection prompts obtain a similar skew reduction without a utility loss. BEND-VLM obtains the lowest skew using gender-labeled FairFace training references. SANER uses predefined gender terms but no per-example attribute labels; it reduces skew further than LATENTRISK-VLM while losing more retrieval utility. This binary-gender, 25-query audit does not establish general demographic debiasing.

Ablation Study

Table 3 shows that semantic residualization is necessary on Waterbirds: removing it lowers average accuracy from 76.3 to 66.6 and WGA from 54.3 to 37.4. Other components are not uniformly beneficial. Removing geometric suppression improves WGA from 54.3 to 57.1, and removing reinjection changes WGA by only 0.1 point. Validation macro accuracy selects $\lambda = 0$ for the full rank-4 editor, making it identical to the no-risk-penalty row. Most importantly, mean contrast outperforms rank four by 2.1 average-accuracy and 6.2 WGA points. The evidence supports a simple direction on Waterbirds, not a general benefit from increasing rank.

Risk Separability Is Insufficient

On the calibration split, the Waterbirds mean-contrast directions separate risk labels with AUCs of 0.977 and 0.932 for landbirds and waterbirds. The CelebA rank-1 probes reach 0.989 and 0.981 for non-blond and blond classes. Yet the edits have opposite WGA effects. These are in-sample diagnostics, not generalization estimates, but they establish an important negative result: making failure linearly separable does not show that the direction corresponds to a hidden group or that removing it will improve group robustness.

Table 1: Classification with frozen CLIP ViT-L/14 encoders, grouped by supervision. WGA, Avg., and Gap are worst-group accuracy, average accuracy, and their difference (percentage points). Attribute use for validation selection counts as attribute supervision.

Supervision	Method	Waterbirds			CelebA		
		WGA	Avg.	Gap	WGA	Avg.	Gap
No example labels	Frozen CLIP	33.6	83.7	50.1	71.5	78.2	6.7
	Prompt ensemble	41.1	83.8	42.7	65.0	77.2	12.1
Task labels only	ERM	29.4	86.6	57.2	16.1	94.3	78.2
	JTT-style calibration	36.1	71.5	35.3	31.1	84.5	53.4
	LATENTRISK-VLM (rank 1)	60.5	78.4	17.9	54.4	91.7	37.3
	LATENTRISK-VLM (rank 4)	54.3	76.3	21.9	52.2	91.6	39.4
Attribute-supervised/selected	SPD (oracle attr.)	22.7	77.2	54.4	67.7	77.8	10.2
	BEND-VLM	31.0	81.2	50.2	58.6	73.5	14.9
	CFR	47.7	74.4	26.7	9.7	39.0	29.3
	CnC	32.9	83.8	50.9	85.3	87.5	2.3
	GIC-Cy + Subsample	58.7	78.8	20.1	28.3	90.6	62.3
	AFR	61.8	80.2	18.3	70.6	88.2	17.7
	LATENTRISK-VLM (attr.-oracle selection)	64.5	80.1	15.6	82.2	85.7	3.5

Table 2: Measured retrieval with frozen CLIP ViT-L/14. MS@100 is mean gender MaxSkew@100 over 25 FairFace stereotype queries (lower is better). Flickr30K reports text-to-image Recall@10 on the Karpathy 1K test split (higher is better).

Method	MS@100 ↓	R@10 ↑
Frozen CLIP	0.355	92.2
Projection prompts	0.326	92.3
BEND-VLM	0.141	91.6
SANER	0.202	85.2
LATENTRISK-VLM	0.323	88.2

Discussion and Limitations

Failure-group alignment is the key assumption. The method never identifies a group as such. It identifies representation variation associated with low margins within a class. Worst-group performance improves only when that variation overlaps the factor defining the worst group and can be removed without erasing class evidence. Waterbirds satisfies this condition strongly enough for rank 1; CelebA does not under task-only selection. Accordingly, LATENTRISK-VLM should be described as latent-risk editing, not unsupervised demographic debiasing.

Discovery and selection are distinct. The CelebA oracle result is informative precisely because it is not a primary

Table 3: Waterbirds ablation with frozen CLIP ViT-L/14. Columns y and a indicate task- and attribute-label use. The last row is not a primary result.

Variant	y	a	Avg.	WGA
Frozen CLIP	N	N	83.7	33.6
SPD (oracle attr.)	N	Y	77.2	22.7
Single mean-contrast direction	Y	N	78.4	60.5
No semantic residualization	Y	N	66.6	37.4
No geometric suppression	Y	N	77.0	57.1
No risk-score penalty	Y	N	76.3	54.3
No neutral reinjection	Y	N	76.3	54.4
Full rank-4 editor	Y	N	76.3	54.3
Rank 4 + attr.-oracle selection	Y	Y	80.1	64.5

result. Group labels are absent from subspace discovery in both rows, yet changing only the validation objective produces a 30.0-point WGA difference. This shows that eliminating group labels from representation discovery does not eliminate the need for a deployment-relevant selection signal. Future work should target label-free robust selection, for example through stability across environments or tail-risk criteria, rather than assuming average accuracy will choose a fair edit.

Attribute supervision remains powerful. CnC outperforms our CelebA oracle under attribute-labeled validation.

The appropriate comparison therefore depends on what is known at adaptation time. Latent-risk editing addresses an unnamed nuisance setting; it should not be presented as a replacement when reliable validation groups are available.

Experimental limitations. All results use one CLIP backbone and one deterministic run, so we cannot claim robustness across architectures or quantify seed variation. The classification datasets are binary, and the retrieval audit reduces gender to the binary labels supplied by FairFace and uses only 25 stereotype queries. The measured rank-1 estimator differs across classification datasets (mean contrast on Waterbirds and a logistic probe on CelebA), so their difference is not a fully controlled estimator comparison. Several baselines are adaptations to frozen CLIP features rather than exact reproductions in their original architectures and data regimes. The same validation split is used for risk discovery and parameter selection, which can favor calibration-specific structure. Finally, a predictive direction is correlational: these experiments do not establish that it causally generates errors or that editing it is safe under unmeasured shifts.

Ethical use. An attribute-free method can still worsen a protected group, as the CelebA result demonstrates. Aggregate accuracy must not be used to certify fairness, and post-hoc subgroup audits remain necessary whenever they are appropriate and lawful. The method should not be deployed to remove demographic information indiscriminately or to infer sensitive attributes. Its appropriate role is as an auditable representation intervention whose group consequences are measured explicitly.

Conclusion

We studied whether frozen-VLM failures can guide representation editing without group annotations. LATENTRISK-VLM uses task-label margins, class residualization, risk-direction estimation, and candidate-wise inference without updating either encoder. It discovers error-related latent risk directions, not sensitive attributes or demographic groups. Rank-1 editing substantially improves Waterbirds WGA, but harms CelebA WGA and trades retrieval utility for a modest FairFace skew reduction. Attribute-selected baselines remain stronger in settings where that supervision is available. Rank four offers no task-only advantage, while attribute-oracle selection exposes a large unresolved selection gap. Failure-derived directions are useful weak supervision when failure and hidden groups align, not a general substitute for group information.

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